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An ecoregion assessment of projected tree species vulnerabilities in western North America through the 21st century

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Running head: Species vulnerabilities in the 21st century

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Abstract

Forest ecosystems across western North America will likely see shifts in both tree species dominance and composition over the rest of this century in response to climate change. Our objective in this paper is to identify which ecological regions might expect the greatest changes to occur. We used the process-based growth model 3-PG, to provide estimates of tree species responses to changes in environmental conditions and to evaluate the extent that species are resilient to shifts in climate over the rest of this century. We assessed the vulnerability of 20 tree species in western North America using the Canadian global circulation model under three different emission scenarios. We provided detailed projections of species shifts by including soil maps that account for the spatial variation in

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soil water availability and soil fertility as well as by utilizing annual climate projections of monthly changes in air temperature, precipitation, solar radiation, vapor pressure deficit and frost at a spatial resolution of 1 km. Projected suitable areas for tree species were compared to their current ranges based on observations at >40,000 field survey plots. Tree species were classified as vulnerable if environmental conditions projected in the future appear outside that of their current distribution $\geq 70\%$ of the time. We added a migration constraint that limits species dispersal to $< 200 \text{ m year}^{-1}$ to provide more realistic projections on species distributions. Based on these combinations of constraints, we predicted the greatest changes in the distribution of dominant tree species to occur within the Northwest Forested Mountains and the highest number of tree species stressed will likely be in the North American Deserts. Projected climatic changes appear especially unfavorable for species in the subalpine zone, where major shifts in composition may lead to the emergence of new forest types.

Introduction

The ecological regions of western North America are geographical areas with diverse environmental conditions and floristic communities. Species that coexist in these ecoregions have distinct adaptations to climate, geology and competing vegetation (Franklin & Dyrness, 1973). Forest ecosystems in western North America have been exposed to increasingly severe drought, a decline in snow cover and lengthening in the growing season, all of which can result in stresses on tree species and permanently alter forest composition (IPCC, 2014; Ganey & Vojta, 2011; Allen et al., 2015; Mote et al., 2005; Waring et al., 2014; Zhou et al., 2001). How vulnerable are tree species in the different ecoregions to the recent changing climate conditions and those projected in the future?

In its latest report, the Intergovernmental Panel on Climate Change (IPCC) states that since the 1950s, unprecedented changes to terrestrial systems have been observed with climate warming (IPCC, 2014). These trends are predicted to continue with elevated greenhouse gas emissions from anthropogenic sources (IPCC, 2014). Increases in average annual air temperature of 3°C are projected by 2080 compared to 1970-1999 (Mote *et al.*, 2010). Temperature increases are predicted to be particularly

acute in northern and interior regions of western North America (Mote *et al.*, 2003) and will far exceed historic rates of change at the end of the last glacial period (Davis & Shaw, 2001). The rapid projected climate change will exceed the capacity of many species to migrate to areas with more favorable conditions (Aitken *et al.*, 2008; Davis & Shaw, 2001). Major changes in species distributions would be accompanied by disturbances caused by fires, insect attacks and diseases that will affect carbon, water and energy balances of affected ecosystems. Recognition of those ecoregions most vulnerable to climate change would permit land managers to concentrate their efforts where protection is most needed.

A number of studies have employed species distribution models to assess the impact of climate change over terrestrial environments (McKenney *et al.*, 2007; Coops & Waring, 2011; Hamann & Wang, 2006; Notaro *et al.*, 2012; Rehfeldt *et al.*, 2012; Jiang *et al.*, 2012). Hamann & Wang (2006) used the Biogeoclimatic Ecological Classification (BEC) system that group distinct forest ecosystems in British Columbia based on common vegetation, climate and site conditions (Meidinger & Pojar, 1991). They combined climate envelope modeling with the BEC system to predict the geographic range of the dominant species of conifers (Hamann & Wang, 2006). McKenney *et al.* 2007 investigated how climate will change the future distribution of 130 tree species in North America. Both studies concluded that species were predicted to expand northward and to higher elevations with losses experienced in their southern ranges. Using similar models in the Southwest U.S., Notaro *et al.* (2012) predicted that the decline in number of species would likely exceed those projected to expand. The predicted declines were expected to be most prominent in high-elevation conifer forests by the end of the 21st century. Rehfeldt *et al.* (2012) predicted an expansion of desert biomes, as well as a northward expansion of temperate and montane forests. Jiang *et al.* (2012) also projected a decline in conifer tree species of up to 24% across western North America especially in the latter half of the 21st century. Forest stands would increasingly be replaced by shrubs and grasslands (Jiang *et al.*, 2012). They highlighted the need to incorporate process-based simulations to understand why species are becoming vulnerable to climate change.

The majority of the above cited studies use empirical approaches to form statistical relationships between current climatic variables and species occurrences and use these to project suitable climate space in the future. However, these relationships are based on the observed distribution of tree species that were established under past climatic conditions that may differ from the climate experienced today. In any case, empirical models form relationships that may not hold in the future, particularly with rising concentrations of atmospheric CO₂ (Pearson & Dawson, 2003; Elith & Leathwick, 2009). An alternative approach is to use process-based models that incorporate biological responses based on experimental data. Process-based models offer a means to predict how tree species might respond to new combinations of environmental conditions (Guisan & Thuiller, 2005). They have the advantage of integrating physiological responses to make predictions of the potential distributions of individual species in the future (Kearney & Porter, 2009; Cuddington *et al.*, 2013). Such models can also provide an understanding on the underlying mechanisms controlling species growth and identify critical environmental limitations. This allows to not only predict how species respond to climate change but also to determine the climate constraints that limit tree growth such as temperature extremes, frost, drought and vapor pressure deficit. As an example, western white pine is limited by soil water shortage conditions and the warmer, drier climate of recent years has led the species to become more susceptible to white pine blister rust leading to increased mortality (Harvey *et al.*, 2008).

A number of process-based models exist that link leaf area index and forest productivity with carbon cycling process and allocation to stems, leafs and roots (Mäkelä *et al.*, 2000; Crookston *et al.*, 2010). Only few of these models have been designed to project both growth and mortality across regional scales (Nightingale *et al.*, 2004) and those that do often assume homogenous patterns in soil water storage (Crookston *et al.*, 2010) and soil fertility (Piao *et al.*, 2013). In this study we use a simplified process-based forest growth model, 3-PG (Physiological Principles Predicting Growth), to infer how tree species are likely to respond to changing climatic conditions. The 3-PG model was developed by Landsberg & Waring (1997) and has been applied and tested on a broad range of forest types around the world (Landsberg *et al.*, 2003). The model provides a balance between complex, fine-scale process-based models and those functioning at annual time-steps. The 3-PG model uses climate and

soil data to determine the extent that environmental limitations can inhibit photosynthesis. Using a hybrid modeling approach we relate these environmental constraints on growth to the recorded presence and absences of the dominant species that exist in the presence of other species. The 3-PG model includes a soil water balance linked with leaf area index, thus incorporating soil factors in the species predictions. The majority of studies currently omit soil properties or hold values constant although it is generally recognized that variation in soil parameters influences tree species growth and distributions and that the inclusion of soil properties can improve model predictions (Syphard & Franklin, 2009; Iverson & Prasad, 1998; Nightingale *et al.*, 2007).

In this study, we incorporate 1-km resolution soil maps depicting available soil water holding capacity and soil fertility (Coops *et al.*, 2012) to provide more realistic representation of spatially diverse site conditions. Using the same model, we previously predicted the current distribution of 20 common tree species across western North America with an average accuracy of 84% (Mathys *et al.*, 2014). We use these species distribution models to extend the previous research by predicting and mapping the vulnerability of ecoregions to climate change in the 21st century. The IPCC defines vulnerability to climate change as a measure of a systems exposure, sensitivity as well as their adaptive capacity to respond to an experienced stress (IPCC, 2014). Our modeling approach accounts for the degree of exposure of the tree species to climate variation and their sensitivity by mapping their tolerance to climate extremes such as drought. We also address the adaptive capacity of the system, how well it responds to the experienced change, by posing limits to the migration rates of a species. To do this, we set a maximum migration distance of 200 m/year based on paleoecological records of dispersal distances and long-term studies on migration rates of different species (Beckage *et al.*, 2008; Coops *et al.*, 2016; Davis, 1989). We acknowledge that the adaptive capacity of a forest ecosystem also depends on genetic adaptation, reproduction, demographic and life history traits. These factors can increase the complexity and uncertainty of projecting tree species responses to climate change. Our focus in this study is to identify areas with suitable climate and soil conditions for species rather than predict actual changes to their distribution in the future. We apply annual, monthly climate data to the end of the 21st century to model suitable sites for tree species using updated climate models in concert

with the most recent IPCC (2014) projections. We predict tree species vulnerabilities in the future by comparing the areas that are projected to become climatically suitable to the current suitable species ranges. Finally, we apply the constraint on the distance that species can migrate to areas with more favorable climate conditions.

Materials and methods

Ecological regions of western North America

Western North America, which spans across Canada and the United States, contains a number of diverse ecoregions that are shaped by unique landforms, soils, climate and vegetation (Franklin & Dyrness, 1973). The major ecoregions include the Marine West Coast Forest, the Northwest Forested Mountains and the North American Deserts (Fig 1).

The Marine West Coast Forest extends from Alaska to Northern California and contains temperate coastal forests. Common tree species found in the ecoregion include Sitka spruce (*Picea sitchensis*), western hemlock (*Tsuga heterophylla*) and Douglas-fir (*Pseudotsuga menziesii*) with western red cedar (*Thuja plicata*), grand fir (*Abies grandis*), Alaskan yellow cedar (*Chamaecyparis nootkatensis*) and coast redwood (*Sequoia sempervirens*). The climate in this region is generally mild with high annual precipitation and this zone supports some of the most productive forests of western North America (Whittaker, 1961).

Tree species that occur in the Interior of Northwest Forested Mountains include Douglas-fir, western hemlock, noble fir (*Abies procera*), western larch (*Larix occidentalis*) and Pacific silver fir (*Abies amabilis*) as well as lodgepole pine, whitebark pine (*Pinus albicaulis*), mountain hemlock (*Tsuga mertensiana*), subalpine fir (*Abies lasiocarpa*), and Engelmann spruce (*Picea engelmannii*) in subalpine regions. This ecoregion has a drier climate than the Marine West Coast Forest and is the second most productive (Waring et al, 2006).

The North American Deserts are located in eastern BC and the southwestern U.S. The predominant coniferous tree species include ponderosa pine, pinyon pine and junipers. The ecoregion has an arid to semi-arid climate and only a 2% forest cover due to the rain shadow caused by the Sierra Nevada and Cascade Mountains (McLaughlin, 1986).

The major ecoregions are further subdivided into more detailed (level III) ecological areas ([http://archive.epa.gov/wed/ecoregions/web/html/na_eco.html#Level III](http://archive.epa.gov/wed/ecoregions/web/html/na_eco.html#Level%20III)), allowing them to be differentiated at a level appropriate for designing regional environmental management strategies (CEC, 1997). The 41 ecoregions included in this study are shown in Fig. 1 and Table 1.

Current and future climate

Climate data required to drive 3-PG includes minimum and maximum air temperature, precipitation, solar radiation, vapor pressure deficit and frost on a monthly basis. The temperature and precipitation values were obtained from ClimateWNA (<http://cfcg.forestry.ubc.ca/projects/climate-data/climatebcwna/>), a program that downscales PRISM (Parameter-elevation Regressions on Independent Slopes Model, dataset Norm71m) data to a 1 km resolution (Wang *et al.*, 2012; Daly *et al.*, 2008). It uses data from weather stations and interpolates them spatially. The downscaling method involves both a bilinear interpolation of the climate data and adjustments of temperature in mountainous terrain. The elevation adjustments are undertaken using a dynamic local regression function that relates monthly temperature with latitude, longitude and elevation (Hamann & Wang, 2005). We used a 90-m Digital Elevation Model (DEM) from the Shuttle Radar Topography Mission (SRTM) and resampled it to 1-km to obtain climate data at a given elevation from the ClimateWNA program. Monthly incoming shortwave radiation was derived by combining the North American Regional Reanalysis (NARR) model with topographic variations (Fu & Rich 2002; Schroeder *et al.*, 2009).

Monthly climatic data averaged from 1976-2009 served as a baseline for this study. We also applied future climate projections to model potential species distributions. Future monthly projections from 2010 through 2100 were obtained from the Canadian Climate Centre's Modeling and Analysis (CCCma) CanESM2 model for three emission scenarios (low (RCP2.6), medium (RCP4.5) and high

(RCP8.5)) as described in the IPCC Fifth Assessment Report (AR5). Although the CCCma model starts with a lower air temperature value during the 20th century and projects more accelerated warming than other GCMs, it is considered as the most appropriate model for the Pacific Northwest Region (Mote *et al.*, 2005). Over the last century, the average air temperature across the region has increased by 0.8 °C. By 2080, the Canadian model projects an additional 3 °C average annual increase and wetter summers, particularly in BC, compared to records from 1970-1999 (Mote *et al.*, 2010; Spittlehouse, 2008). These data were incorporated in the 3-PG model to evaluate how environmental stresses might affect the distribution of tree species in the future.

Modeling approach

We first ran 3-PG to obtain predictions on the seasonal constraints to species growth. Details on the modeling approach are described in Mathys *et al.* (2014) and Coops *et al.* (2009). Briefly, the 3-PG model combines climate data with species response parameters to infer the extent that climatic and soil constraints inhibit photosynthesis. The resulting monthly climate modifiers include responses to soil water and atmospheric vapor pressure deficits, suboptimal air temperatures and frost, normalized on a continuous scale from 0 (most limiting) to 1 (no constraints on growth). Monthly baseline climate (1976-2009) drove the model, from seedling establishment to a stand age of 50 years. Spatial variations in available soil water holding capacity and soil fertility were included (Mathys *et al.*, 2014; Coops *et al.*, 2012). The next stage in model development was to create a decision tree that combined the climate modifiers derived from 3-PG with recorded occurrences of the 20 tree species (DBH>2.54cm) on 43,404 field plots distributed throughout the study area (Table 2). These data were acquired from Forest Inventory and Analysis (FIA) plots by the US Forest Services in the United States, from vegetation resource inventory plots in British Columbia, Canada as well as from species occurrence data provided by the Alberta Environment and Sustainable Resource Development (Schroeder *et al.*, 2009; Bechtold & Patterson 2005). The decision tree model ranked the relative importance of the climate modifiers to predict both the presence and absence of tree species based on their recorded distribution. The models predicted species occurrence and absence with an overall accuracy of 84% and a kappa statistic of 0.79 (Mathys *et al.*, 2014). To expand the analysis, we

compared the availability of suitable habitats in the future using climate data averaged from 2075-2100 with that projected under baseline conditions (1976-2009). We also estimated within the current range of a tree species, where it was likely to experience unfavorable conditions (defined at a vulnerability threshold of 70% for the years sampled) using monthly averaged climate data for each year from 2075 to the end of the 21st century. Similarly, we identified potential areas for range expansion by mapping where favorable conditions might occur $\geq 70\%$ of the time.

To provide more realistic predictions of potential migration, we imposed a maximum rate of dispersal of 200 m per year (Coops *et al.*, 2016) on tree species expansion based on paleobotanical records of species migrations (Davis, 1989; Beckage *et al.*, 2008). While a more mechanistic understanding of the constraints on species migration would be desirable, the additional detail required would likely increase the uncertainty of the model predictions (Thuiller *et al.*, 2008). Many studies chose to omit using species dispersal distances due to limited data available on the rate that species might disperse (Parmesan & Yohe, 2003), the extent the landscape has become fragmented or experienced a change in land use (Pearson & Dawson, 2005; Opdam & Wascher, 2004). Choosing a simple approach allows us to address species migration rates as a measure of adaptive capacity by accounting for the distances they can potentially migrate.

Within each ecoregion, we calculated the percentage change in the future suitable locations of 20 tree species and contrasted their current range (1976-2009) with that predicted between 2075 and 2100. We calculated the number of species stressed under changing climate by ecoregion (EPA level 3) to determine those forested areas in western North America highly vulnerable to projected shifts in climatic conditions. To do this we counted the number of pixels classified as stressed for all species and averaged the values across each ecoregion unit. We only considered ecoregions that currently have at least three conifer species present to eliminate those ecoregions that contained a high diversity of species that were excluded from this study (Fig. 1). Among the 20 dominant conifer tree species, we focused on the vulnerability of two of the most widely distributed, Douglas-fir and lodgepole pine.

We also chose whitebark pine, because although its distribution is more limited, it is a keystone species in the subalpine zone providing an important food source for grizzly bears and contributing to watershed stability (Burns & Honkala, 1990).

Results

Fig. 2 compares the current distribution of three tree species Douglas-fir, lodgepole pine and whitebark pine as defined from 1976-2009 to their projected suitable habitat from 2075-2100 using the most optimistic emission scenario that simulates the lowest GHG emissions (RCP2.6). The tree species are well distributed throughout the study region, with all three currently present in the Northwest Forested Mountains. Douglas-fir also occurs in the Marine West Coast Forest, with its range extending from British Columbia to California and further in the southwest. We projected that in another century, both lodgepole pine and whitebark pine will have much more constricted ranges in the Northwest Forested Mountains. Douglas-fir on the other hand, is likely to respond favorably to the projected climate conditions with an expansion throughout this and other ecoregions.

The decision tree models developed for Douglas-fir, lodgepole pine and whitebark pine identify the relative importance of environmental constraints in determining the presence and absence of the tree species (Fig. 3). Accordingly, Douglas-fir is present on sites with favorable air temperatures during the spring and limited by drought in the fall. Unfavorable high temperatures in the summer appear to limit both lodgepole pine and whitebark pine. Whitebark pine has a relative high tolerance to frost in the summer (<0.78) and lodgepole pine is tolerant to frost conditions below 0.41 in the fall. Lodgepole pine also appears to depend on sufficient soil water recharge in the wintertime, associated with an accumulation of a snowpack in the winter. Soil water deficits imposed an important environmental constraint in two of the tree decision tree models in Fig. 3.

Ecoregions with the largest projected decline in the distribution of dominant species are shown in Table 2. The ecoregions with the greatest species vulnerability are mostly located within the Northwestern Forested Mountains. These include the following ecoregions: the Watson Highlands (6.1.5), the Arizona/New Mexico Plateau (10.1.7), the Southern Rockies (6.2.14) and the Idaho

Batholith (6.2.15). The potential reduction in species distributions ranged from a 1% to 64% (Table 3). Both lodgepole pine and western white pine showed the greatest decline in their predicted future distributions, whereas subalpine fir and Douglas-fir were most likely to become stressed within the ecoregions.

The areas where individual tree species were projected to expand or contract their range under different emission scenarios are shown in Fig. 4. Douglas-fir displayed increased stress in the southern part of its distribution within the Northwest Forested Mountains and North American Deserts in the United States (Fig. 4a). Areas within the northern part of its distribution were projected to become favorable for species expansion, both in the Marine West Coast Forest along the coast of BC as well as on the Canadian side of the Northwest Forested Mountains in BC and Alberta. For the RCP 2.6 scenario, Douglas-fir had the lowest vulnerability ranking, maintaining 78% of its current range (1976-2009). The RCP 8.5 scenario would, according to our model, cause up to a 59% reduction in the current range of Douglas-fir whereas the RCP 4.5 scenario projected the greatest area for expansion (22%).

Both lodgepole pine and whitebark pine were predicted to decline throughout their entire current distribution under all emission scenarios (Fig. 4b and d). Areas deemed unsuitable for lodgepole pine ranged from 67% (RCP 2.6) to 81% (RCP 8.5) of its current range. Sitka spruce, a species in close proximity to the Pacific Coast throughout the Marine West Coast Forest, is likely to be more limited in the southern part of its range in Washington, Oregon and California, while likely to expand northwest along the coast of BC under future environmental conditions (Fig. 4c). 80% of its current range remained suitable for the species under the RCP 2.6 scenario. The RCP 8.5 scenario projected both the greatest stress (39%) on Sitka spruce in its current range and most potential for range expansion (9%). Other species within the study region showed a variety of responses to future climatic projections (not shown). Many subalpine species were projected to become increasingly stressed where they occur within the Northwest Forested Mountains. These include mountain hemlock and subalpine fir, which both are predicted to decline throughout their current ranges, although subalpine fir should remain present in parts of BC, Alberta and the Rocky Mountains in the US. While

ponderosa pine may become highly stressed where it occurs within the North American Deserts, within the Northwest Forested Mountains some areas favored its expansion. Species such as western hemlock, western red cedar and Alaska yellow cedar retained their habitat within the Marine West Coast Forest but were predicted to experience increased stress in parts of the Northwest Forested Mountains.

The highest number of species were characterized as likely to be stressed within the North American Deserts (Fig. 5). The Thompson-Okanagan Plateau (10.1.1) ecoregion was projected to have the highest vulnerability in the study area, with an average of six species stressed for all emission scenarios (66% of the original number of species present). Other ecoregions with high stress values included the Wyoming Basin (10.1.4), the Columbia Plateau (10.1.2), the Northern Basin and Range (10.1.3), the Colorado Plateaus (10.1.6), the Snake River Plain (10.1.8), the Central Basin and Range (10.1.5) and the Arizona/New Mexico Plateau (10.1.7) with averages of five species predicted as stressed in the future and relative values of 54-70%. Ecoregions within the Marine West Coast Forests appeared to be the least vulnerable to projected climate shifts from 2075-2100 compared to their current occurrence in 1976-2009. These include the Coast Range (7.1.8) and Coastal Western Hemlock-Sitka Spruce Forests (7.1.5) where less than 35% of the species present were predicted to be vulnerable to future climate conditions.

Discussion

Our projections of tree species vulnerabilities to climate shifts varied among ecoregions. Over the coming century, the largest number of species stressed among the 20 included in the analysis, were predicted to occur in the North American Deserts to the east of the Sierra Nevada and Cascade Mountain ranges in the western United States and British Columbia, Canada. During the growing season, this ecoregion experiences high air temperatures and limited water supply, especially in the southern areas of the ecoregion (CEC, 1997). With projected future climate change, tree species that currently inhabit this environment will likely be reduced to those that can adapt to even more extreme conditions. The majority of tree species we analyzed that currently occupy the desert ecoregion were

projected to be subject to increased stress, although ponderosa pine and Utah juniper were shown to retain suitable habitat in some areas. A subset of the desert ecoregion, the Thompson-Okanagan Plateau in the Interior of BC, was projected to have the highest number of species stressed. With the climate there already among the warmest and driest in BC, endemic forest disturbances such as drought, wildfire and insect attacks are also set to increase in frequency and intensity (Haughian *et al.*, 2012; Wheaton, 2005). In relative terms, the desert ecoregion ranked among the top ten in terms of species stress and ecoregions within the Northwest Forested Mountains experienced a high relative vulnerability as well.

Within the Northwest Forested Mountains we projected a range of responses to occur. Some species became increasingly stressed while others remained resilient and may even expand into new habitat that becomes climatically suitable. We found subalpine species such as lodgepole and whitebark pines to be especially vulnerable to climate change, with both species showing overall range contractions. Species such as Douglas-fir, meanwhile, responded with both a contraction and expansion in their distribution within the Northwest Forested Mountains. The relative species stress in this ecoregion was as high as within the North American Deserts, indicating that a large proportion of original species present are vulnerable in both major ecoregions.

The Marine West Coast Forest, an ecoregion with a wet, mild climate contributing to the highest forest productivity in the study area, displayed the lowest species vulnerability. Although the Marine West Coast Forest climate is projected to change the least among the ecoregions studied, increased precipitation could lead to more flooding and slope failures (Haughian *et al.*, 2012). The projected distribution of Douglas-fir was maintained throughout most of the Marine West Coast Forest with a potential of expansion northward in British Columbia. Conditions remained suitable for Sitka spruce as well, except in the southern part of the ecoregion in Oregon and Washington where the species is projected to experience increased stress.

One of the benefits of the 3-PG model is that it allows us to not only predict where increased species vulnerabilities are likely but also to identify the environmental changes that cause stress (Landsberg *et al.*, 2003). For instance, our decision tree models ranked spring temperature as the most limiting constraint on growth of Douglas-fir. Douglas-fir is a species known to be well adapted to both the mild, wet climate of the Marine West Coast Forest as well as to drier conditions in the Northwest Forested Mountains, though it is still vulnerable to excessively warm and dry conditions (Klinka *et al.*, 1999). Our results also showed that Sitka spruce was most limited by frost conditions during the fall. It grows well in waterlogged soils but has a low tolerance to both extreme temperatures and frost and does not tolerate drought well (Klinka *et al.*, 1999). This relatively shade tolerant species has a good chance to dominate sites unsuitable for other species (Burns & Honkala, 1990).

In the Northwest Forested Mountains ecoregion, our decision tree model ranked summertime temperatures as the most limiting modifiers for lodgepole pine, indicating that the projected increase in air temperatures with climate change would be the chief stressor on the species. Soil water conditions in the winter and frost in the fall also affected predicted lodgepole pine distribution. In recent years, an increase in both winter temperatures and summer drought have led to conditions that increased the vulnerability of lodgepole pine to the mountain pine beetle outbreak that has killed an unprecedented number of trees in western Canada and the United States (Carroll *et al.*, 2006; Safranyik & Wilson, 2006).

Whitebark pine, meanwhile, had a highly variable response to altered climate conditions, with our results showing that the most important climate constraint on whitebark pine was summer temperature. The current decline of the whitebark pine throughout its distribution is of great concern and government agencies have devised conservation strategies to help protect and restore the species in both Canada and the United States (Haughian *et al.*, 2012; Hansen & Phillips, 2015). Concern over whitebark pine decline is in part due to its role as a keystone species, providing an important food source for animals and facilitating forest growth in alpine habitats (Burns & Honkala, 1990). Whitebark pine has been affected by both white pine blister rust and the mountain pine beetle, with those trees weakened by summer drought and inhabiting high altitudes most affected (Tomback &

Resler 2007; Haughian *et al.*, 2012). Similar to our findings, other studies project a continued decline of whitebark pine over the next century with climate warming (Schrag *et al.*, 2008; Chang *et al.*, 2014). Climate-induced disturbances such as an increase in the frequency and size of fires as well as changing species competitors may also impact whitebark pine in Montana (Loehman *et al.*, 2011). Knowledge of the climate and soil factors that stress whitebark pine as detailed in this study can help to develop conservation strategies to reduce the threats of climate change.

Our projections agree with other studies that predict increased stress of species in the southern part of their ranges coupled with potential range expansions to higher latitudes and altitudes (McKenney *et al.* 2007; Iverson & Prasad, 1998; Hamann & Wang, 2006). Similar to our findings, a number of studies that have reported increased species die-off in the North American Deserts in recent years, although these studies generally focus on areas within the Southern USA rather than considering the full range of tree species (Ganey & Vojta, 2011; Mueller *et al.*, 2005; Moore *et al.*, 2015). Recent tree mortality has been observed in the mixed-conifer forests of Arizona that are part of the North American Deserts ecoregion (Ganey & Vojta, 2011). Ma *et al.* (2015) similarly reported that in Australia, the greatest vulnerability of ecosystems to drought occurs in semi-arid regions. In general, it is projected that there will be an overall expansion of desert biomes while temperate forest regions will expand northward, becoming replaced by shrublands in their southern ranges (Rehfeldt *et al.*, 2012; Jiang *et al.*, 2012). An analogous trend was also predicted in the mountains of Colorado and Utah, where conifer species would become partially replaced by grasslands except at the highest elevations where tree cover was expected to increase (Notaro *et al.*, 2012). Studies elsewhere in the world also indicate increased vulnerability of subalpine species to climate warming (Thuiller *et al.*, 2005; Rehfeldt *et al.*, 2012; Hansen & Phillips, 2015). Indeed, an overall decline in suitable habitat for subalpine species such as whitebark and lodgepole pine has been projected by the end of the 21st century in a number of studies that used bioclimate envelope models (Bell *et al.*, 2014; Gray & Hamann, 2013; Rehfeldt *et al.*, 2012).

Unlike many of the above empirical studies, the 3-PG model allowed us to identify the underlying climate constraints that cause species vulnerabilities. In the case of subalpine species, warmer winters, earlier snowmelt and drier summers all contribute to tree species stress. Soil properties also have a well-recognized influence on the growth and distribution of tree species though they have still seen a limited inclusion in species distribution models (Syphard & Franklin, 2009; Iverson & Prasad, 1998; Mathys *et al.*, 2014). By incorporating spatially variable soil information in our species models, we gain additional knowledge on areas that are suitable for range expansion and where we can expect stress to increase a species' vulnerability. The soil water modifier was an important environmental constraint in our study that helped account for much of the variation in the majority of our decision tree models: among the 20 conifer species evaluated in this study, 75% responded to changes in soil water holding capacity, with up to a 27% change in their distribution when varying this property (Mathys *et al.*, 2014).

Adding spatial constraints to the rate of migration provides a more realistic picture of the implications of rapid changes in climate. It is clearly unrealistic, without human assistance, to accommodate the conclusion that many tree species would need to migrate 100-250 km per decade to keep up with the rate of projected climate warming (McKenney *et al.* 2007; Iverson & Prasad, 1998; Hamann & Wang, 2006), a distance that exceeds historic postglacial migration rates during the Quaternary (Aitken *et al.*, 2008; Davis & Shaw, 2001). In recognition of the inability of species to migrate rapidly, foresters are considering assisting the movement of species into more suitable areas, while expanding, where possible, their genetic diversity (Hoegh-Guldberg *et al.*, 2008; McLachlan *et al.*, 2007). The limits we impose on species migrations, therefore, help to classify locations that are suitable for natural range expansion. The suitable habitat of some of the tree species can be 50-90% lower when migration is constrained compared to an assumption of uninhibited migration (Coops *et al.*, 2016). Curtailing the rates and distances of tree species migration allowed us to identify areas that are not only projected to become climatically suitable for each of the species assessed, but also where species are unlikely able to reach from their current locations. These results can help to focus assisted migration efforts on suitable locations that are beyond the natural dispersal limits of tree species.

Our predictions of future tree species distributions can serve as guidelines to distinguish areas where we may expect increased stress to occur from those that are more resilient to future climate change. The extent of the distributions themselves should not be assumed to be accurate due to uncertainties surrounding different climate model projections, the rate of disturbances, habitat fragmentation and the availability of seed sources (Corlett & Westcott, 2013). The extent that tree species ranges will change depends on complex, interactive processes such as species interactions, genetic adaptation, CO₂ fertilization effects, disturbance regimes and land use change. The effects of land cover change varies by ecosystem type, with coastal species and those in early successional stages being more adapted to human-caused habitat fragmentation than slower growing, more shade tolerant species (Opdam & Wascher, 2004). A better understanding of the environmental thresholds that cause tree mortality would improve our ability to recognize species that are most vulnerable to climate change (Allen *et al.*, 2015; Park *et al.*, 2014). Changes in snowpack are also important and are currently only indirectly deduced with the 3-PG model. Days with low temperatures and precipitation that exceeds losses through evapotranspiration are indicators of a heavy snowpack accumulating in certain locations. These sites can be identified in the decision tree models as they rarely experience a high vapor pressure deficit, low soil moisture or temperatures favorable for growth in the wintertime. These conditions are especially important for subalpine species, such as lodgepole and whitebark pines that normally occur on sites with the presence of a heavy snowpack in the winter. Despite these limitations, by predicting how tree species might respond to environmental change at annual time steps, we may be able to gain additional insights regarding the importance of extreme events such as drought and flooding. Knowledge on the ecoregions that are most vulnerable to climate shifts enables land managers to concentrate their efforts to provide refugia and to assist tree species migration. In the ecoregions where increased stress is most likely, it may be advisable to plant tree species that are better able to persist in a future climate. And finally, increasing habitat connectivity can help species migrate to new areas that are projected to favor their growth (Jongman *et al.*, 2011).

Our model predictions serve as a basis to predict where the potential for tree migration is strongest. Imposing limits on the rates that species can migrate provides insights into their ability to respond to climate change naturally and where forest management practices may be necessary to assist their migration. Our study also provides mechanistic understandings of the climate and soil constraints that affect species vulnerabilities thus allowing forest managers to better devise adaptation strategies to climate change. We projected the greatest number of species will become stressed in the North American Deserts and the greatest vulnerability of dominant species to occur in the Northwest Forested Mountains. A decline of these species will free sites for the invasion of tree species that are better adapted to changing climate conditions. In other ecoregions we can also expect changes in species community composition with new forest types emerging as some species become more abundant than others (Williams *et al.*, 2004). While the projections remain to be confirmed, they provide a framework for understanding where to focus conservation efforts to build forest resilience and which species are likely to persist in a future climate.

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Table 1 Level III Ecoregions within study region of western Canada and United States

Ecoregion III	Name
10.1.1	Thompson-Okanogan Plateau
10.1.2	Columbia Plateau
10.1.3	Northern Basin and Range
10.1.4	Wyoming Basin
10.1.5	Central Basin and Range
10.1.6	Colorado Plateaus
10.1.7	Arizona/New Mexico Plateau
10.1.8	Snake River Plain
11.1.3	Southern and Baja California Pine-Oak Mountains
13.1.1	Arizona/New Mexico Mountains
5.4.1	Mid-Boreal Uplands and Peace-Wabaska Lowlands
5.4.2	Clear Hills and Western Alberta Upland
6.1.4	Wrangell and St. Elias Mountains
6.1.5	Watson Highlands
6.1.6	Yukon-Stikine Highlands/Boreal Mountains and Plateaus
6.2.1	Skeena-Omineca-Central Canadian Rocky Mountains
6.2.10	Middle Rockies
6.2.11	Klamath Mountains
6.2.12	Sierra Nevada
6.2.13	Wasatch and Uinta Mountains
6.2.14	Southern Rockies
6.2.15	Idaho Batholith
6.2.2	Chilcotin Ranges and Fraser Plateau
6.2.3	Columbia Mountains/Northern Rockies
6.2.4	Canadian Rockies
6.2.5	North Cascades
6.2.6	Cypress Upland
6.2.7	Cascades
6.2.8	Eastern Cascades Slopes and Foothills
6.2.9	Blue Mountains
7.1.4	Pacific Coastal Mountains
7.1.5	Coastal Western Hemlock-Sitka Spruce Forests
7.1.6	Pacific and Nass Ranges
7.1.7	Strait of Georgia/Puget Lowland
7.1.8	Coast Range
7.1.9	Willamette Valley
9.2.1	Aspen Parkland/Northern Glaciated Plains
9.3.1	Northwestern Glaciated Plains
9.3.3	Northwestern Great Plains
9.4.1	High Plains
9.4.3	Southwestern Tablelands

Table 2 The 20 tree species and number of presence and absence plots utilized in this study

Species	Latin name	Presence	Absence	% Present	% Absent	Total (n)
Lodgepole pine	<i>Pinus contorta</i>	9236	34168	21.3	78.7	43404
Douglas fir	<i>Pseudotsuga menziesii</i>	12959	30445	29.9	70.1	43404
Piñon pine	<i>Pinus edulis</i>	1387	42017	3.2	96.8	43404
Subalpine fir	<i>Abies lasiocarpa</i>	5577	37827	12.8	87.2	43404
Engelmann spruce	<i>Picea engelmannii</i>	4476	38928	10.3	89.7	43404
Whitebark pine	<i>Pinus albicaulis</i>	1161	42243	2.7	97.3	43404
Western larch	<i>Larix occidentalis</i>	1489	41915	3.4	96.6	43404
Pacific silver fir	<i>Abies amabilis</i>	1119	42285	2.6	97.4	43404
White fir	<i>Abies concolor</i>	2178	41226	5.0	95.0	43404
Grand fir	<i>Abies grandis</i>	1816	41588	4.2	95.8	43404
Noble fir	<i>Abies procera</i>	218	43186	0.5	99.5	43404
Utah juniper	<i>Juniperus osteosperma</i>	1738	41666	4.0	96.0	43404
Sitka Spruce	<i>Picea sitchensis</i>	374	43030	0.9	99.1	43404
Western white pine	<i>Pinus monticola</i>	826	42578	1.9	98.1	43404
Alaska yellow cedar	<i>Callitropsis nootkatensis</i>	740	42664	1.7	98.3	43404
Western red cedar	<i>Thuja plicata</i>	2992	40412	6.9	93.1	43404
Western hemlock	<i>Tsuga heterophylla</i>	3083	40321	7.1	92.9	43404
Mountain hemlock	<i>Tsuga mertensiana</i>	773	42631	1.8	98.2	43404
Ponderosa pine	<i>Pinus ponderosa</i>	5701	37703	13.1	86.9	43404
Quaking aspen	<i>Populus tremuloides</i>	6190	37214	14.3	85.7	43404

Table 3 Percent and absolute changes in the potential distribution of dominant species by ecoregion from 2075-2100 compared to their current range

Ecoregion III	Name	Species	Change in distribution	
			Percent change (%)	Net change (km ²)
9.3.3	Northwestern Great Plains	Western white pine	-64	-47367
6.1.5	Watson Highlands	Lodgepole pine	-64	-75938
10.1.7	Arizona/New Mexico Plateau	Ponderosa pine	-58	-17550
6.2.14	Southern Rockies	Subalpine fir	-35	-28063
6.2.13	Wasatch and Uinta Mountains	Subalpine fir	-22	-4882
6.2.15	Idaho Batholith	Douglas-fir	-19	-10130
6.2.10	Middle Rockies	Subalpine fir	-8	-8396
10.1.4	Wyoming Basin	Douglas-fir	-1	-590
6.2.2	Chilcotin Ranges and Fraser Plateau	Subalpine fir	-1	-779

Fig. 1 Ecological regions (level III) in western Canada and United States as defined by the U.S. Environmental Protection Agency. The major ecoregions (level I) are outlined in bold.

Fig. 2 Current and future potential suitable habitat of lodgepole pine, Douglas-fir and whitebark pine from (a) 1976-2009 and (b) 2075-2100 under the RCP2.6 emission scenario.

Fig. 3 Decision tree models of Douglas-fir, lodgepole pine and whitebark pine ranking the relative importance of four environmental constraints that determine the presence and absence of tree species (ranked between 0 and 1). Temp = temperature modifier.

Fig. 4 Areas deemed unsuitable (purple) or suitable (green) for (a) Douglas-fir, (b) lodgepole pine, (c) Sitka spruce and (d) whitebark pine under three future emission scenarios in 2075-2100 compared to their current distributions (grey) in 1976-2009.

Fig. 5 Predicted number of species stressed throughout the study regions for the three emission scenarios (a) RCP2.6, (b) RCP4.5 and (c) RCP8.5 in 2075-2100.























